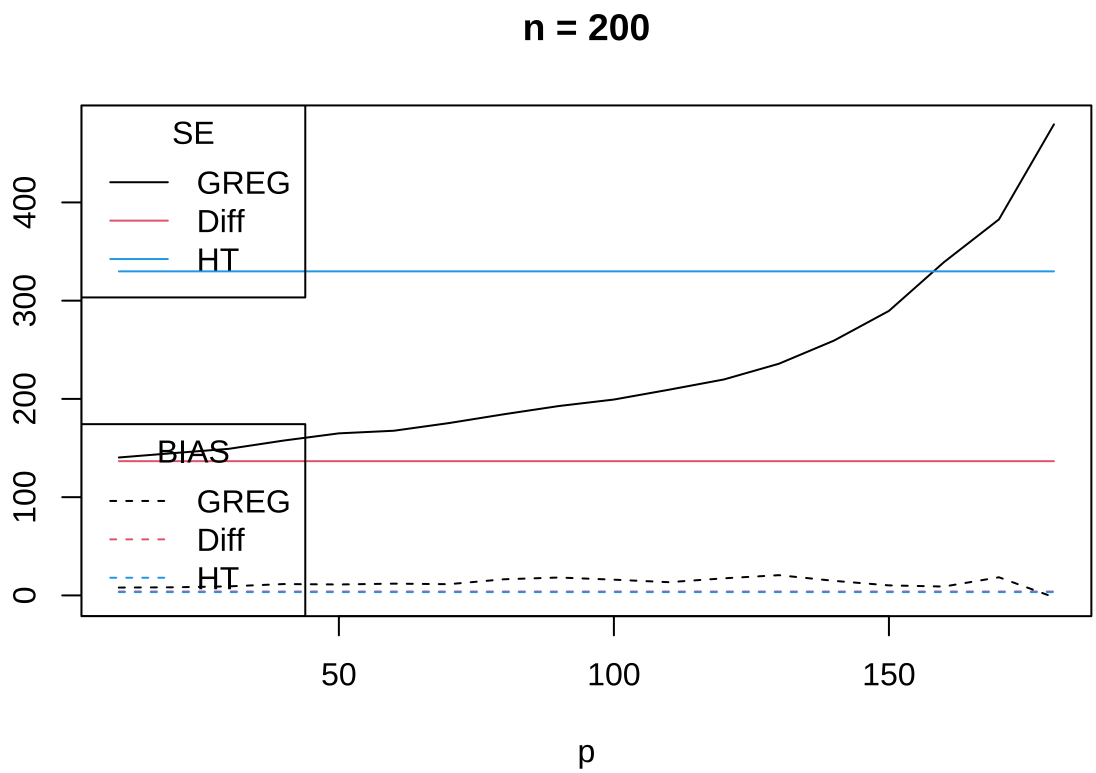
High dimensioanl simulation study

```r
N = 2000
n = 200
BIAS_res = NULL
SE_res = NULL
pvec = 1:18 * 10

X_whole = matrix(rnorm(N * 500, 2, 1), nr = N, nc = 500)

for(p in pvec){
  # p = 10
  set.seed(1)
  print(p)
s = 5

X = X_whole[,1:p]
beta = c(rep(1, s), rep(0, p - s))
e = rnorm(N, 0, 1)
y = X %*% beta + e
t_y = sum(y)

# pi1 = X[,1]^2 / sum(X[,1]^2) * n
pi1 = rep(n / N, N)
d1 = 1 / pi1

res = NULL
res2 = NULL
res3 = NULL
SIMNUM = 500
for(simnum in 1:SIMNUM){
  # set.seed(simmum)
  Index = sample(1:N, size = n, replace = FALSE, prob = pi1)
  y_s = y[Index]
  X_s = X[Index, ]
  d1_s = d1[Index]
  lm_obj = lm(y_s ~  X_s, weights = d1_s)
  beta_hat = lm_obj$coefficients[-1]
  #drop(solve(t(X_s) %*% diag(d1_s) %*% X_s, t(X_s) %*% diag(d1_s) %*% y_s))

  # beta_hat = unname(lm(y_s ~  t(apply(X_s, 1, function(k) k
  # - colSums(X_s * d1_s) / N)), weights = d1_s)$coefficients[-1])

  y_HT = sum(y_s * d1_s)

  y_diff = drop(colSums(X) %*% beta + sum((y_s - drop(X_s %*% beta)) * d1_s))

  y_GREG = drop(colSums(X) %*% beta_hat + sum((y_s - drop(X_s %*% beta_hat)) * d1_s))
```

```
  y_model = drop(colSums(X) %*% beta_hat)

  # sigma_Debiased = sqrt(drop(t(e_s)  %*% Omega %*% e_s * N^2 / n^2))

  res = rbind(res, c(HT = y_HT, Diff = y_diff, GREG = y_GREG))


}

BIAS = colMeans(res - t_y)
SE = apply(res, 2, function(x) sqrt(var(x) * (length(x)-1)/length(x) ))
RMSE = apply(res - t_y, 2, function(x) sqrt(mean(x^2)))

cbind(BIAS, SE, RMSE)

BIAS_res = rbind(BIAS_res, BIAS)
SE_res = rbind(SE_res, SE)
}
```

```
## [1] 10
## [1] 20
## [1] 30
## [1] 40
## [1] 50
## [1] 60
## [1] 70
## [1] 80
## [1] 90
## [1] 100
## [1] 110
## [1] 120
## [1] 130
## [1] 140
## [1] 150
## [1] 160
## [1] 170
## [1] 180
```

```
plot(pvec, SE_res[,3], type = "l", col = 1, ylim = c(min(BIAS_res), max(SE_res)), main = sprintf("n = %
points(pvec, SE_res[,2], type = "l", col = 2)
points(pvec, SE_res[,1], type = "l", col = 4)
legend("topleft", c("GREG", "Diff", "HT"), col = c(1,2,4), lty = 1, title = "SE")

points(pvec, BIAS_res[,3], type = "l", col = 1, lty = 2)
points(pvec, BIAS_res[,2], type = "l", col = 2, lty = 2)
points(pvec, BIAS_res[,1], type = "l", col = 4, lty = 2)
legend("bottomleft", c("GREG", "Diff", "HT"), col = c(1,2,4), lty = 2, title = "BIAS")
```



n = 200
SE
GREG
Diff
HT
BIAS
GREG
Diff
HT
0
100
200
300
400
50
100
150
p